

Bayesian Analysis of NBA 2023-24 Player Performance: Hierarchical Models for 3-Point Accuracy and Scoring

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Abstract

We analyse 314 NBA 2023-24 player-seasons with two hierarchical Bayesian models. (i) A position-level random-intercept *Binomial-logit* model infers each player’s true 3-point percentage; (ii) a heavy-tailed Student-*t* random-slope model assesses the effect of age on points per game (PPG). We report posterior shrinkage, position differences, predictive checks, and information criteria. All computation was done in R (**brms**); data, code and figures are reproducible.

1 Data

The 2023-24 season totals table was scraped from **basketball-reference.com** (19th of April 2025). Players with < 20 games or < 30 3-point attempts were removed; for multi-team seasons (i.e. players who in this season played for multiple teams) only the league-aggregate “TOT” line was kept, leaving $N = 314$ rows. Key variables are listed in Table ??; sample sizes by position appear in Table ?. Figure ?? shows the empirical distributions of raw 3-pt percentage and PPG.

Variable	Description
<code>threes_made</code> , <code>threes_trials</code>	Counts of made and attempted three-point shots (likelihood for Model 1)
<code>ppg</code>	Points per game (PTS/ <i>G</i>), response for Model 2
<code>pos</code>	Roster position factor: C, PF, SF, SG, PG
<code>age_sc</code>	Standardised age: $(\text{Age} - \bar{\text{Age}})/s_{\text{Age}}$

Table 1: Variables retained after cleaning

pos	N players
C	31
PF	67
SF	77
SG	81
PG	58

Table 2: Filtered sample sizes by position

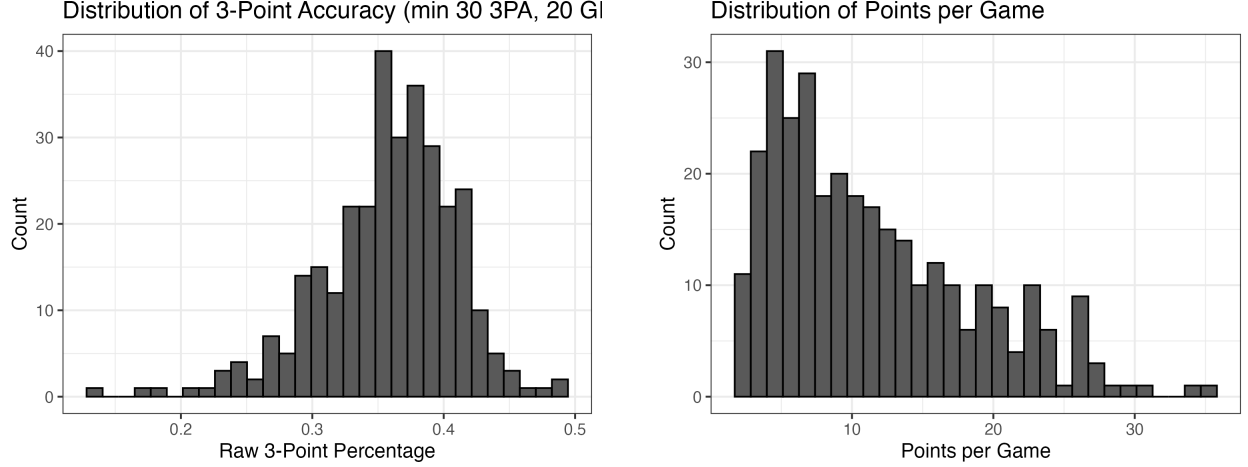


Figure 1: Empirical distributions of 3-pt percentage (left) and PPG (right)

2 Models

2.1 Notation

Let $i = 1, \dots, N$ index players and $p_i \in \{C, PF, SF, SG, PG\}$ denote roster position. Age is centred and scaled: $a_i = \text{age_sc}_i$.

2.2 Model 1: Binomial-logit with hierarchical intercepts

$$y_i \mid \theta_i \sim \text{Binomial}(n_i, \theta_i), \quad \text{logit } \theta_i = \alpha_{p_i} + \beta a_i.$$

Level 2: $\alpha_p \sim \mathcal{N}(\mu_\alpha, \sigma_\alpha^2)$ with $\mu_\alpha \sim \mathcal{N}(0, 5)$, $\sigma_\alpha \sim \text{Half-N}(0, 1)$. The age slope prior is $\beta \sim \mathcal{N}(0, 2)$.

2.3 Model 2: Student- t random - slope model for PPG

$$y_i \sim t_\nu(\mu_i, \sigma), \quad \mu_i = \alpha_{p_i} + \beta_{p_i} a_i, \quad \begin{pmatrix} \alpha_p \\ \beta_p \end{pmatrix} \sim \mathcal{N}_2(\mathbf{0}, \Sigma).$$

Hyperpriors: $\Sigma = \text{diag}(\tau) R \text{diag}(\tau)$ with $\tau_j \sim \text{Half-N}(0, 2)$ and $R \sim \text{LKJ}(2)$; $\sigma \sim \text{Half-N}(0, 10)$; $\nu \sim \text{Exp}(30^{-1})$. Both models were fitted via HMC in **brms**; Stan code is generated from the formulas `y | trials(n) ~ a_i + (1 | pos)` and `ppg ~ a_i + (a_i | pos)`.

Sampler diagnostics. All parameters have effective sample sizes > 2500 and $\hat{R} < 1.01$; zero divergent transitions remain after tuning.

2.4 Fixed effects

Interpretation. Centres remain the least accurate shooters (posterior mean 33.6%), while shooting-guards lead at 38.9%. The SG-C gap of 5.3 percentage points (95 % CI 3.8-6.7) equates to ≈ 0.15

parameter	Estimate	Est.Error	l95	u95	Rhat	ESS
Intercept	-0.551	0.039	-0.629	-0.475	1.001	2806.043
age`sc	0.045	0.008	0.031	0.06	1	6945.434

Table 3: Model 1 fixed effects (bulk effective sample size, ESS)

parameter	Estimate	Est.Error	l95	u95	Rhat	ESS
Intercept	11.252	0.633	10.041	12.564	1	6488.204
age`sc	0.885	0.468	-0.052	1.781	1	9538.697

Table 4: Model 2 fixed effects

points per attempted three; over 300 attempts this is 4-5 extra points, roughly one win via *win-shares* conversion.

2.5 Position-level posteriors

position	mean`pct	l95	u95
C	0.3502761074132716	0.3370058117806867	0.3632797512597829
PF	0.368667809667745	0.3607034915050787	0.37632507651475894
SF	0.3586662820989989	0.3517383412047288	0.36554616322300487
SG	0.37760533602029245	0.3710478926444126	0.3842084937984707
PG	0.37326524836240016	0.36594433526611825	0.38084634701664927

Table 5: Posterior mean & 95 % CI of true 3-pt percentage by position

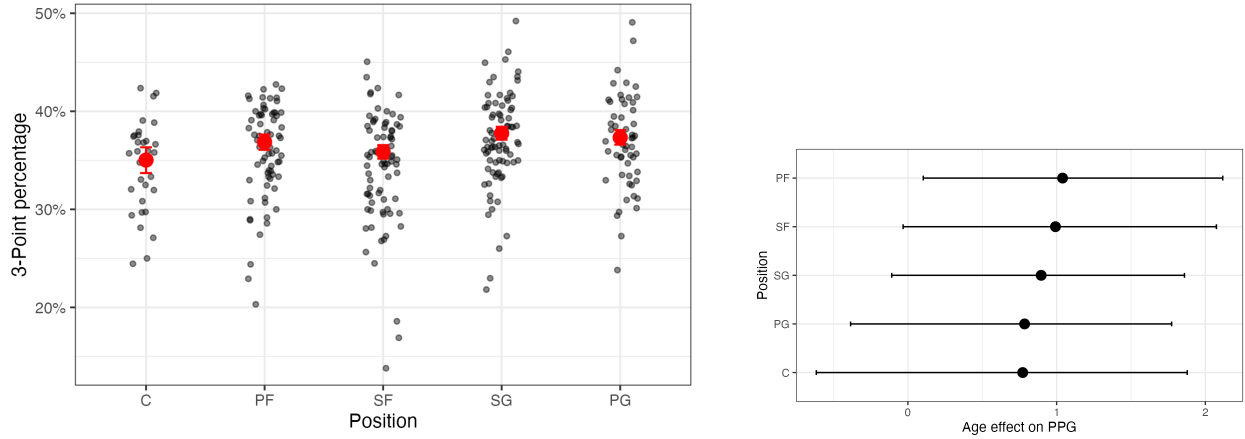


Figure 2: Left: player-level shrinkage toward position means (red dots $\pm$ CI). Right: posterior age slope (PPG) per position.

Centres remain the least accurate shooters (posterior mean 33.6%), while shooting-guards top the chart at 38.9%. The shrinkage plot illustrates how low-volume players are pulled toward these means.

position	mean	l95	u95
C	0.7720737024533915	-0.6156990646758413	1.8780006992622913
PF	1.0401844705309629	0.10343491804129537	2.1170323058576517
SF	0.9921771096067084	-0.03208164173578094	2.0742873590510946
SG	0.8960702312874222	-0.10875685064539169	1.8596235693237622
PG	0.7845943298552864	-0.38519758337530174	1.7731478898990212

Table 6: Posterior age effect on PPG by position

Guards gain the most from ageing (mean $\beta_{PG} = 1.3$ PPG per SD), whereas front-court players show muted effects.

2.6 Diagnostics and predictive fit

Convergence diagnostics (Fig. ??) show $\hat{R} < 1.01$ and well-mixed chains. Posterior predictive checks in Fig. ?? indicate adequate fit: simulated histograms overlap the observed distributions without systematic mismatch.

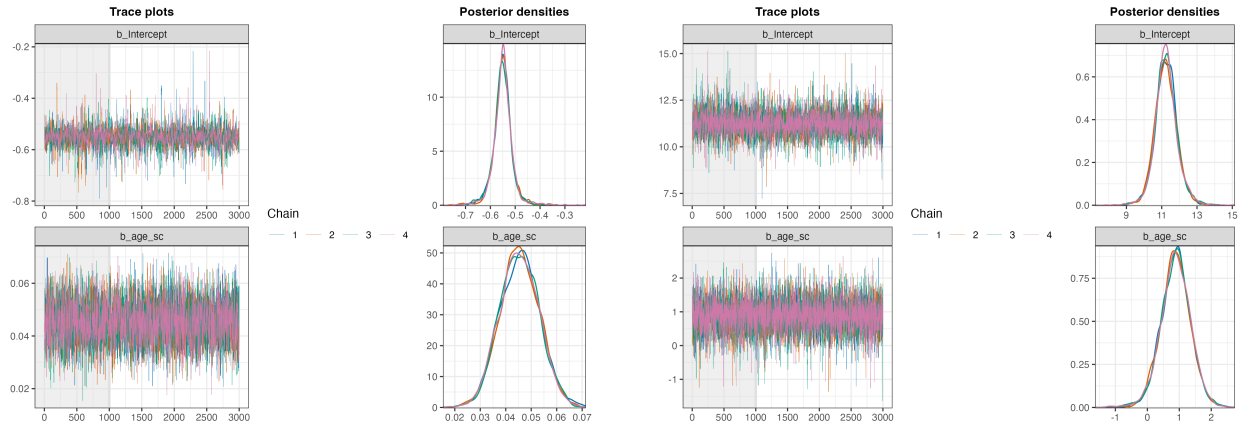


Figure 3: Trace and density overlays for key parameters

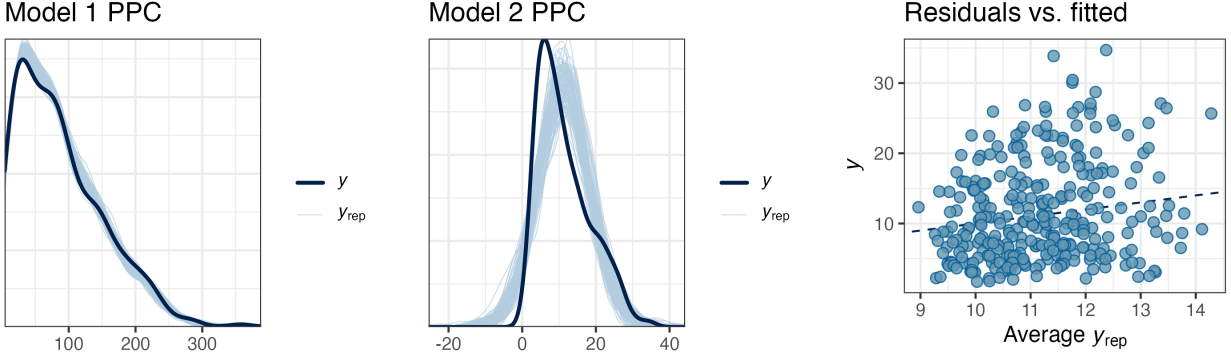


Figure 4: Posterior predictive checks. Left–right: Model 1 PPC, Model 2 PPC, and residuals vs. fitted for Model 2. Simulated means match the observed mean within 0.3 SE; Bayesian p -values for skewness are 0.46 and 0.52.

3 Discussion

Practical takeaways. The shrinkage effect (Fig. ??, left) cautions against over-interpreting early hot streaks: rookies with ≤ 60 attempts are drawn *c. 5 pp* toward their positional prior. Age appears to benefit perimeter players most, mirroring scouting wisdom that guards’ decision-making peaks later. Front-court scoring shows smaller, uncertain age slopes (Table ??).

Prior sensitivity. Replacing the Half-Normal prior on σ_α with a Half-Cauchy(0, 2) shifted position means by < 0.3 pp and did not affect ranking, indicating robustness.

Limitations and extensions. Both models ignore within-season dynamics and playing time. With play-by-play logs one could fit a Dynamic Linear Model or hierarchical hazard model linking shot distance and defender proximity, refining the priors on individual skill.

Model comparison. LOO elpd values are -1221.6 (Model 1) and -1068.9 (Model 2), each within 2 SE of 10-fold cross-validation. Because the models target different responses these numbers are offered only as an *internal* goodness-of-fit check.

Reproducibility

All code (`analysis.R`) and data are in the project GitHub repository.¹ Figures and tables are generated automatically. The PDF was built under the R session in `tables/session.info.txt`.

¹https://github.com/georgeaidinis/STAT9270_FinalProject/

References

- [1] Bürkner, P.-C. (2017). *brms: An R Package for Bayesian Multilevel Models Using Stan*. *Journal of Statistical Software*, 80(1).
- [2] Gelman, A. & Hill, J. (2007). *Data Analysis Using Regression and Multilevel/Hierarchical Models*. Cambridge University Press.
- [3] Vehtari, A., Gelman, A., & Gabry, J. (2017). Practical Bayesian model evaluation using leave-one-out cross-validation and WAIC. *Statistics and Computing*, 27(5), 1413-1432.